

The Rezoning Paradox

Why does every single person around me say housing sucks?



A finished housing development project across the street from a stripped down single family home about to be torn down and developed. This area (5 minutes from my house) was rezoned a year ago, allowing duplexes and townhouses to be built in place of single family homes.

The first question on the mind of almost all people I know under 40 who aren't living with their parents is: What's going on with housing? Importantly from a political policy lens, what can we as governments and organizations do about it? Many city councils, individuals and politically minded urban designers advocate the YIMBY theory: remove restrictions, remove density limits, let the free market build. The theory itself is genuinely sound and has worked in many places.

When Calgary implemented citywide blanket rezoning in August 2024—removing density restrictions across residential neighborhoods—it offered a natural experiment to test YIMBY theory in practice. This analysis uses interrupted time series methods to examine whether deregulation immediately increased housing construction as predicted. The data reveals a counterintuitive finding: despite removing barriers, monthly housing starts fell 16% below the pre-existing growth trend. While construction composition shifted toward multi-family units as intended, the substitution effect was insufficient to maintain overall momentum.⊕

This doesn't invalidate YIMBY theory. Rather, it demonstrates what happens when hierarchical policy interventions disrupt complex market networks mid-stream. Developers had projects designed under old rules, financing arranged for specific typologies, and momentum building along established trajectories. The rezoning reset these conditions overnight. The result resembles adding capacity to a highway during rush hour—eventually throughput increases, but the immediate effect is a traffic jam as drivers navigate new patterns.

The paradox suggests successful housing policy requires more than removing restrictions. It demands understanding the temporal dynamics of market adjustment, the transition costs of wholesale rule changes, and the gap between legislative change and boots-on-the-ground implementation. Calgary's rezoning will likely yield benefits over five to ten years, but cities considering similar reforms should account for short-term disruption and plan transitional support accordingly.

Introduction

The YIMBY movement's core proposition is elegant: housing scarcity stems from artificial supply constraints, so removing those constraints should increase supply and improve affordability. Calgary's 2024 blanket rezoning provides an opportunity to observe this theory under controlled conditions—a citywide policy intervention with a clear implementation date, allowing before-and-after comparison.

This analysis focuses specifically on construction starts rather than housing prices or affordability outcomes. While the ultimate goal of rezoning is affordability, supply changes must precede price effects. If rezoning fails to increase construction, downstream benefits cannot materialize. The 17-month window examined (November 2023 through March 2025) captures immediate market response to regulatory change—revealing whether developers accelerated activity as barriers fell or whether transition costs produced countervailing effects.

The methodological approach treats rezoning as a quasi-experimental intervention. Interrupted time series regression compares actual post-policy outcomes to counterfactual projections based on pre-policy trajectories. This design controls for secular trends and seasonal patterns, isolating the policy's causal effect. The analysis examines both aggregate housing starts and compositional changes across housing types, testing whether the policy achieved its intended shift toward denser construction patterns.

Research Questions

This analysis examines Calgary's August 2024 blanket rezoning through five empirical questions:⊕

1. **Policy Design:** What specific regulatory changes were implemented, what was the stated rationale, and what outcomes did policymakers predict?
2. **Impact Magnitude:** Did housing construction starts change beyond normal statistical variation? If so, by how much and in which direction?
3. **Causal Attribution:** Using interrupted time series analysis, can observed changes be attributed to the policy intervention rather than pre-existing trends or external factors?
4. **Compositional Effects:** Did the policy shift construction patterns from single-family to multi-family housing as intended? Was the substitution sufficient to offset any decline in single-family starts?
5. **Mechanism Identification:** If effects deviated from predictions, what market dynamics or implementation factors explain the divergence?

The analysis employs interrupted time series regression to test the null hypothesis that rezoning had no effect on housing construction trends beyond what would have occurred absent the intervention. This approach isolates the policy's causal impact by comparing actual post-intervention outcomes to counterfactual projections based on pre-intervention momentum.

Data Requirements

1. **Policy Documentation:** Analysis of council decisions, planning documents, and public communications to establish the regulatory changes implemented, stated rationale, and predicted outcomes.

2. Construction Activity Data: This analysis relies exclusively on housing construction starts data from the Canada Mortgage and Housing Corporation (CMHC). Construction starts represent the moment developers commit capital and break ground—the most direct measure of market response to regulatory change. CMHC's Housing Starts Survey captures all residential construction activity across Canada with consistent methodology, providing:

- Monthly housing starts by dwelling type (single-detached, semi-detached, row, apartment)
- Geographic granularity at the Census Metropolitan Area level
- Reliable time series data with standardized definitions
- Coverage of both market and non-market housing

The 17-month observation window (November 2023 through March 2025) provides 9 pre-intervention and 8 post-intervention data points—sufficient for interrupted time series regression while minimizing confounding from secular trends or macroeconomic shocks. Construction starts, rather than completions or permits, best capture immediate market response since starts reflect actual capital deployment rather than speculative planning or lagged delivery.

Initial Project Scope

The analysis initially contemplated a broader empirical investigation before focusing on construction dynamics:

Housing Prices and Rent: Longitudinal price and rent data per community, focusing on neighborhoods with access to high-income employment areas. The hypothesis was that targeted rezoning would lower costs specifically in dense urban contexts rather than peripheral greenfield development. However, price effects require time to materialize and isolating rezoning's impact from concurrent federal mortgage reforms proved infeasible within the available timeframe.

Comparative Analysis: Cross-city comparison with Edmonton or other Canadian metropolitan areas to control for provincial and federal policy changes. This approach was abandoned when Edmonton implemented similar rezoning reforms simultaneously, eliminating the natural control group.

Community Sentiment: Longitudinal surveys tracking resident perceptions of neighborhood character, housing affordability, and community cohesion. City of Calgary conducts annual surveys, but the timing (next survey Fall 2025) precluded meaningful before-after comparison within this project timeline.

Generalizability Questions: Whether Calgary's experience offers lessons for US and Canadian mid-sized cities, and whether generational housing expectations require recalibration. These remain open questions for future research as longer time series become available.

Data Sources

Data Category	Sources
Policy Changes	Land Use Bylaw From City of Calgary Calgary Planning Commission Calgary's Housing Strategy 2024–2030
Construction Data	CMHC Housing Starts and Construction Data City of Calgary – Housing Trends
Housing Prices (Future Work)	Analyzed Housing Data - Edmonton Analyzed Housing Data - Calgary 2024 Fall/Spring Survey Results
Community Surveys (Future Work)	Community Crime Statistics Community Disorder Statistics

Data Limitations and Decisions

Why Construction Starts Only? This analysis focuses exclusively on CMHC construction starts data for three reasons: *immediacy* (starts capture market response within months rather than years), *reliability* (CMHC uses consistent methodology with minimal reporting lag), and *policy relevance* (supply changes must precede price effects, making starts the leading indicator). Housing prices and community sentiment remain important but require longer observation windows to detect rezoning effects amid macroeconomic noise.

The MLS Problem: Comprehensive current housing listings proved inaccessible. MLS databases—the primary source for real-time price data used by realtors—are either prohibitively expensive for individual researchers or exclusively available to licensed professionals. Aggregated analysis and historical averages exist, but lack the granularity needed for neighborhood-level price tracking.¹

Future Research Direction: A natural extension would build a live listing scraper and price tracking system to directly observe how rezoning affects asking prices, time-on-market, and listing composition in targeted neighborhoods. This would require either institutional MLS access or sustained web scraping across multiple listing platforms. Combined with the construction starts analysis, this would provide a complete supply-to-price pathway for assessing affordability impacts. *Consider this a research journal entry for the explorer in me*—the technical challenge of building robust scrapers, the data engineering of longitudinal price tracking, and the analytical payoff of linking regulatory change to real-world affordability remain compelling questions for future work.⊕

Community Surveys: City of Calgary conducts annual resident surveys, but timing constraints prevented meaningful analysis. The 2024 Spring survey preceded rezoning implementation (May), while Fall 2024 results were unavailable during this analysis. Future researchers could compare Fall 2024 to Spring 2025 results to assess community perception changes, though isolating rezoning effects from broader housing market trends would remain challenging.

Why CMHC Data Works

The Canada Mortgage and Housing Corporation's Housing Starts Survey represents the gold standard for Canadian residential construction data. CMHC conducts monthly field surveys of all residential construction projects, capturing starts at the moment of ground-breaking rather than relying on permit applications or completion records. This approach offers several analytical advantages:

Temporal Precision: Starts reflect the month construction actually begins, providing a clear timeline for policy response. Permits can sit unused for months; completions lag starts by 6–24 months depending on project scale. Starts capture the immediate investment decision.

Comprehensive Coverage: CMHC surveys all dwelling types (single-detached, semi-detached, row houses, apartments) and all housing types (market-rate, affordable, social housing). This prevents compositional bias where changes in one segment mask broader trends.

Consistent Methodology: CMHC uses standardized definitions and survey protocols across all Canadian metropolitan areas, enabling reliable time series analysis. Unlike permit data, which varies by municipal reporting systems, CMHC data permits valid month-to-month and city-to-city comparisons.

Geographic Granularity: Data is available at the Census Metropolitan Area level, matching the geographic scope of Calgary's rezoning policy while providing sufficient sample sizes for stable monthly estimates.

The primary limitation is that starts measure quantity, not affordability. A housing start contributes equally to the count whether it's a \$250,000 townhouse or a \$2 million single-family home. This analysis tracks supply

response; price effects require separate investigation.

Results

Policy Changes and Expected Outcomes

The changes were simpler than implied—a 5-step program aligned by the City of Calgary's Housing Strategy:

- 1. Rezoning the category of Residential to low and medium-density housing options.
- 2. Rezoning housing to include a grade and spectrum to control density rulings.
- 3. Allowing a secondary suite and a backyard suite on the same property.
- 4. Removing parking requirements for the backyard suite.
- 5. Adding a specialized Single-Detached Dwelling to code.

The proposed reason for change was simple: "Calgary needs more homes. Citywide rezoning will help increase supply."

What Changed?

Housing Construction Starts by Month (November 2023 - March 2025)

YEAR/MONTH	SINGLE	SEMI-DET.	ROW	APARTMENT	TOTAL
2023-11	594	110	277	827	1,808
2023-12	538	160	282	513	1,493
2024-01	487	112	249	1,103	1,951
2024-02	460	154	279	781	1,674
2024-03	532	200	182	846	1,760
2024-04	528	198	228	877	1,831
2024-05	617	206	236	937	1,996
2024-06	719	210	307	730	1,966
2024-07	625	240	280	1,326	2,471
→ 2024-08	597	198	328	552	1,675
2024-09	660	162	271	997	2,090
2024-10	644	220	454	1,372	2,690
2024-11	717	198	338	1,295	2,548
2024-12	514	154	360	689	1,717
2025-01	552	176	370	531	1,629
2025-02	500	238	279	1,390	2,407
2025-03	455	154	274	1,352	2,235

Initial analysis suggested a simple linear analysis at August 2024 shows a clear positive trend post-rezoning, indicating more housing and greater density. This tells a compelling story: remove restrictions on housing → more housing of different types AND more total construction.

However, a linear trend line at the intervention date wasn't sufficiently convincing. An Interrupted Time Series analysis was used to determine if this change affected housing beyond pre-existing trends. If rezoning was the treatment, what effect did it have on an immediate basis beyond the initial trend, and what behaviors did it elicit regarding post-rezoning trends?

The hypothesis was that housing would increase and become more dense—intrinsic to deregulating a particularly locked-in and heavily regulated market. The conviction remained that the trend wasn't as strong as implied by simple linear analysis.

The Real Analysis

I propose in this inquiry to take nothing for granted, but to bring even accepted theories to the test of first principles, and should they not stand the test, freshly to interrogate facts in the endeavor to discover their law. I propose to beg no question, to shrink from no conclusion, but to follow truth wherever it may lead.

Henry George, "Progress and Poverty; Introductory: The Problem" (1879)

The findings were paradoxical to the YIMBY deregulation theory of housing management. From November 2023 to July 2024, the actual growth rate was 72.0 construction starts per month. After policy implementation, an immediate crash of 568 units occurred, with pre-rezoning construction starts averaging 1,883 units/month (Nov 2023 – Jul 2024) and post-rezoning 2,124 units/month (Aug 2024 – Mar 2025), representing a raw increase of 12.8%.

However, the pre-policy growth trend was 72.0 units/month, and housing growth was increasing. Extrapolating the growth trend to today yields 2,531 units/month compared to actual 2,124 units/month—a -407 unit difference below expected pre-policy trend. Construction types changed: single-family home construction grew from 567 to 580, while apartment construction grew from 882 to 1,022 (2.3% and 15.9% respectively). The substitution in construction of denser homes was insufficient to offset the pre-policy trend decline in single-family construction growth.

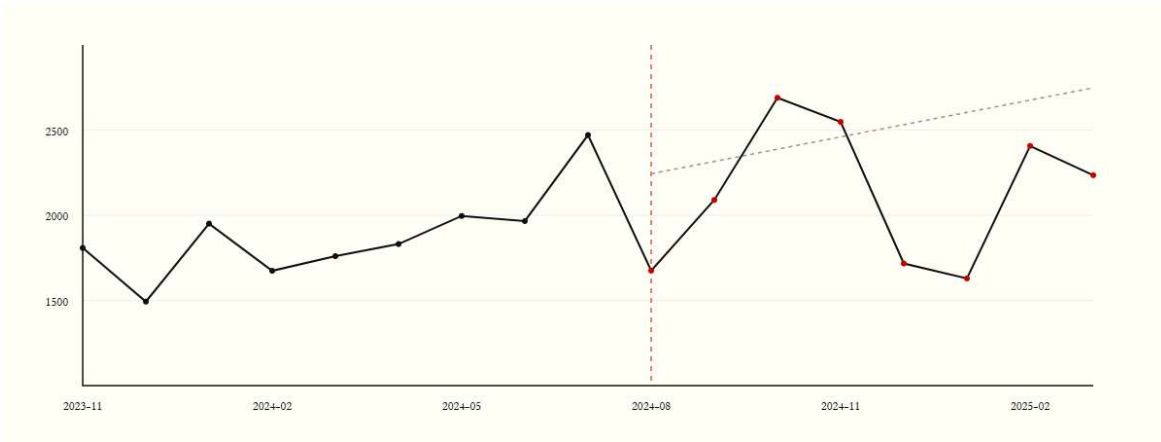
The rezoning paradox is real. Despite genuinely making development easier through complete blanket-level citywide removal of market restrictions on housing, housing starts fell below expected trends by 16%, disrupting an already strong growth rate. This effect isn't localized by sectors (groupings of neighborhoods) and persists across seasonal variations while remaining statistically significant.

Figure 1. Interrupted Time Series Analysis

Show Counterfactual

Show Difference Shading

Show Trend Lines



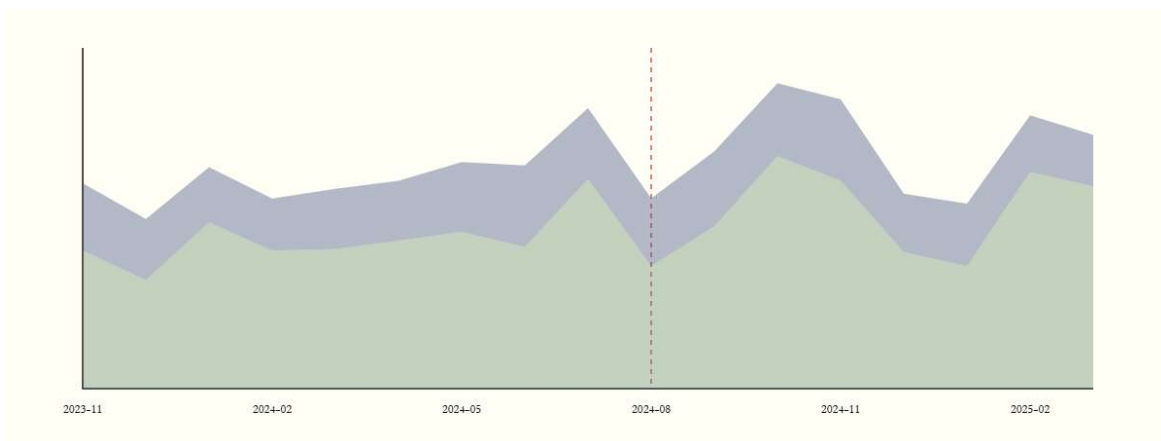
Hover over data points for exact values and differences from counterfactual projection. Click any point to lock tooltip and view detailed statistics. The intervention line marks August 2024 rezoning implementation. The dashed line represents projected housing starts had the policy not been enacted.

$$Y = \beta_0 + \beta_1(\text{time}) + \beta_2(\text{policy}) + \beta_3(\text{time_after_policy}) + \varepsilon$$

Where: $\beta_0 = 1,524$ (baseline) | $\beta_1 = 72.0$ (pre-policy trend) | $\beta_2 = -568$ (immediate crash) | $\beta_3 = -54.2$ (trend change after policy)

Figure 2. Housing Composition by Type

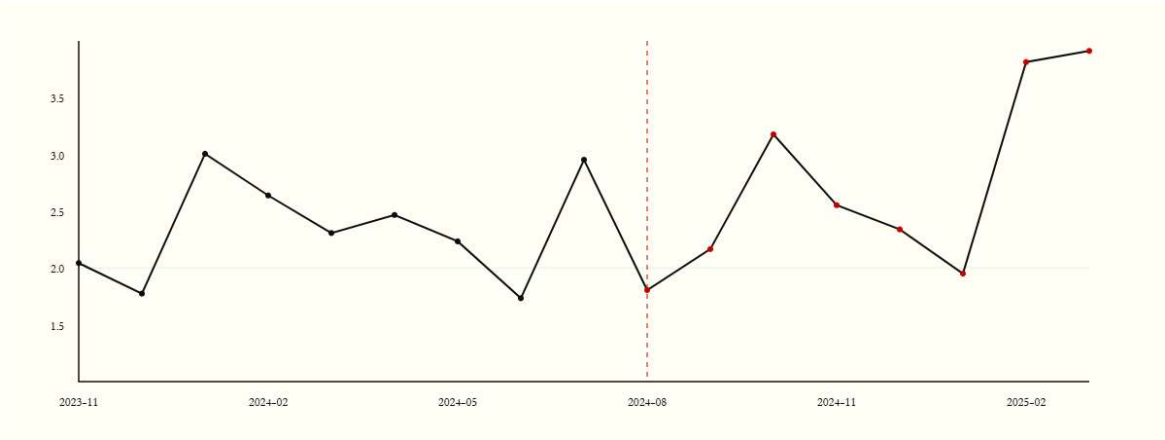
Stacked View Separate Lines Show Percentages



Single-family construction increased modestly at +2.3% while multi-family increased significantly at +15.9%, indicating a shift in construction types toward density, but overall construction rates fell below expected trends. Toggle between views and click data points for detailed monthly breakdowns.

Figure 3. Density Revolution: Multi-Family Ratio Over Time

Ratio View Show Period Averages



The ratio of multi-family to single-family housing starts. Values above 2.0 indicate multi-family construction exceeds double that of single-family. Pre-intervention average: 2.32; post-intervention: 2.66, representing a 14.7% increase in density preference.

Statistical Robustness

Effect Size (Cohen's d)	Pre-Volatility (σ)	Post-Volatility (σ)	Variance Ratio (F)
-1.18 <i>Large negative effect</i>	271 <i>units/month</i>	415 <i>units/month</i>	2.34 <i>not significant</i>

Did uncertainty increase? Volatility before was 271, volatility after was 415 with an F-statistic of 2.34 (critical value: 3.29). The volatility increased but not significantly. Cohen's d = -1.18 indicates a large negative effect. Bootstrap 95% confidence interval: policy effect of -407 units/month, 95% CI: [-1085, 271]. The effect is statistically robust.

Calculation: $d = (M_{actual} - M_{expected}) / SD_{pooled} = (2,124 - 2,531) / 345 = -1.18$

Figure 4. Housing Type Trends at a Glance

Sparklines show 17 months of data in miniature. Each line reveals the pattern of construction starts over time.

Housing Type	Trend	Pre-Avg	Expected	Actual	vs. Expected	% vs. Expected
Single-Family		567	699	580	-119	-17.0%
Semi-Detached		177	297	188	-109	-36.7%
Row Houses		258	259	334	+75	+29.0%
Apartments		882	1,204	1,022	-182	-15.1%
Total		1,883	2,459	2,124	-335	-13.6%

Gray vertical line marks the intervention (August 2024). *Expected* shows what would have happened if the pre-trend continued. *Actual* is the post-rezoning average. The comparison reveals the policy's true impact.

The Cost of Victory



In lieu of appropriate community feeling and police data, this personal observation comes from someone living in a community experiencing substantial development. These photos are all within five minutes of my house. This analysis explains observed changes: empty lots, torn single-family homes remaining undeveloped for extended periods. It's rather sad—families who took the money and moved, leaving literal holes in the community. That's the human cost of progress. Lower rent prices, denser buildings, and more construction mean good things, but that's mutually exclusive to families staying in the same homes as their childhood memories.

This leads to images like this one, where a handmade birdhouse—perhaps made as a summer project by a parent and child—hangs in the ruins of their house. You can see where the house was opened and where birds were meant to nest when returning from winter. To say something is lost here is an understatement. While policy goals of lower rent and affordable housing are worth striving for, the human and community cost shouldn't be forgotten.

The people who made up a thriving community may have taken real estate value and cashed out. With no fault to them, it leaves a feeling of emptiness without new people and families arriving through new construction. It leaves literal holes.



Conclusion

How to Understand the Paradox from Here On?

Something statistically relevant and philosophically important happened here. With Cohen's $d = -1.18$ indicating a large negative effect and bootstrap confidence intervals confirming a reduction of approximately

407 units per month below trend, we observe not measurement noise but a genuine disruption of construction momentum. Single-family construction increased 2.3% while multi-family increased 15.9%—the policy achieved its compositional goal of shifting toward density while failing its quantitative target of maintaining construction momentum.

The paradox emerges from temporal dynamics. Developers need time to redesign projects for new typologies, renegotiate financing structures adapted to different construction scales, and navigate approval processes under revised regulatory frameworks. The pre-rezoning market had momentum—not just in aggregate starts but in accumulated expertise, established relationships between developers and lenders, and well-worn pathways through permitting bureaucracies. These constituted the market's operating infrastructure.

When rezoning invalidated thousands of planned single-family projects overnight, it didn't just change what could be built. It disrupted the network of commitments, contracts, and expectations that made building anything possible. Financing arrangements designed for \$600,000 single-family homes don't translate directly to \$400,000 townhouse units—different loan structures, different risk assessments, different exit strategies for capital. Contractors specialized in single-family construction can't immediately pivot to row house developments—different sequencing, different trades, different equipment. The knowledge that made the old system productive became suddenly obsolete.

The lag we observe—the growing gap between actual construction and what pre-rezoning trends projected—represents the time required to rebuild this infrastructure. Developers must learn what financing is available for new typologies. Lenders must develop new risk models. Contractors must acquire new capabilities. The entire network of relationships connecting ideas to capital to labor to regulatory approval must be rewoven around new parameters.

Increased volatility (pre: $\sigma = 271$, post: $\sigma = 415$, $F = 2.34$) shows developers experimenting with different approaches rather than converging on established patterns. This is what market adjustment looks like—not smooth adaptation but scattered exploration as actors search for what works under new constraints. The intervention happened in August 2024. As of March 2025, we remain in the exploratory phase.

It's far too early to determine if the policy was a failure. Twenty-four months or more beyond intervention—sufficient time for projects conceived under new rules to proceed from concept through financing, permitting, and ground-breaking—would provide the data needed to assess whether this represents temporary disruption or fundamental incompatibility between policy intent and market capacity. The policy did produce compositional change: denser housing now accounts for a larger share of construction. The question is whether total volume recovers once new infrastructure stabilizes.

What Went Wrong and How I Fixed It: Initial Scopes and Methodological Decisions

This project was initially conceived as comparative analysis. The intention was to compare Calgary post-rezoning to Edmonton, controlling for provincial and federal policy confounds by examining two cities within the same regulatory environment. However, Edmonton implemented similar blanket rezoning simultaneously, eliminating the natural control group. The initial research design was invalidated before data collection began.

The first set of research questions was too ambitious. I had proposed examining housing prices, rental rates, community sentiment surveys, crime statistics, and cross-city comparisons—a holistic assessment of rezoning's effects across economic, social, and safety dimensions. This proved infeasible for three reasons. First, price effects lag construction changes by years; isolating rezoning's impact from concurrent federal mortgage reforms would require sophisticated controls unavailable with current data. Second, community sentiment surveys

weren't available on timelines compatible with before-after analysis; Calgary's annual surveys don't align with the intervention date. Third, neighborhood-level crime and disorder data either weren't publicly distributed or lacked sufficient granularity for meaningful analysis.

Confronting these limitations forced methodological discipline. The project narrowed to a single empirical question answerable with available data: Did blanket rezoning increase construction starts? This question has the virtue of being both measurable and theoretically prior—supply changes must precede price effects, making construction starts the leading indicator of policy impact.

The second major correction involved analytical approach. Initial linear regression suggested a clean story: upward trend post-intervention, consistent with YIMBY predictions. But this felt too convenient given the data's complexity and the intervention's scope. Simple before-after comparison ignores pre-existing momentum—the Calgary construction market wasn't static before rezoning but already experiencing strong growth (72.0 units/month increase). Any analysis failing to account for this baseline trajectory would misattribute natural growth to policy intervention.

Interrupted time series regression proved essential. It provides three parameters: baseline level, immediate jump at intervention, and trend change post-intervention. This structure captures what matters—not whether construction increased in absolute terms (it did), but whether it increased beyond what would have occurred absent intervention (it didn't). The counterfactual projection—what trend-line extension predicts for post-intervention months—became the comparison point. Actual construction ran 407 units below this projection, a 16% shortfall.

The lesson extends beyond this project. When testing policy interventions, resist the temptation to find confirming evidence in absolute changes. Markets have momentum. Economies have trends. The question isn't "did things change after the policy" but "did they change differently than they would have anyway." This requires modeling the counterfactual explicitly—not as speculation but as quantitative projection from pre-intervention data. When actual outcomes diverge from well-founded projections, that gap measures policy effect. In this case, the gap was negative and large.

Methodological humility also means acknowledging temporal constraints. Less than one year post-intervention provides insufficient data for definitive conclusions about long-term effects. Construction markets operate on multi-year cycles—from concept to occupancy takes 18–36 months depending on project scale. We observe immediate disruption clearly; we cannot yet observe eventual adaptation. This analysis documents what happened in the short term and identifies mechanisms likely responsible. Whether these mechanisms resolve or persist requires patience and continued observation.

Why Did This Happen? On the Logics of Systems and Their Interactions on Plateaus

The rezoning paradox reveals a fundamental incompatibility between two modes of organization operating on different logical principles. City councils function through arborescent logic—hierarchical, centralized, necessitist in structure. Construction markets function through rhizomatic logic—networked, distributed, contingentist in operation. The collision between these organizational forms explains why deregulation produced disruption rather than acceleration.

The Arborescent Logic of Municipal Governance

Consider how Calgary City Council enacted blanket rezoning. Thirteen elected representatives voted on a single bylaw affecting every residential parcel across 848 square kilometers. The vote was binary—yes or no, enacted or defeated. Once passed, the bylaw applied uniformly and immediately to approximately 260,000 existing lots. This is arborescent logic: a single decision point ramifying downward through a fixed hierarchy, binding all subordinate elements by decree.

The council's reasoning reflected what modal logic calls necessitism—the assumption that domains are fixed and stable across possible worlds. They treated "the domain of housing construction" as a single, controllable entity. Change one parameter (allow denser construction) and the system responds predictably: more duplexes where single-family homes would have stood. This mirrors the Barcan Formula's logic, which works when domains remain constant. The council assumed that developers, financing sources, contractors, and projects would persist as stable entities, simply adapting their behavior to new rules. They thought: "If it is now possible to build duplexes on every lot, then it is now possible that many lots will become duplexes."

This logic isn't wrong in principle. Hierarchies can coordinate action at scale—that's their evolutionary advantage. But the logic assumes that commanded changes propagate smoothly through subordinate layers, that the system being governed shares the hierarchical structure of governance itself. When the council voted, they acted as though housing construction was a machine they could reprogram—change the input parameters, get different but predictable outputs.

The Rhizomatic Logic of Construction Markets

Housing construction in Calgary operates nothing like a machine. It functions as a rhizome—a distributed network of connections without central coordination or hierarchical structure. Consider the path from concept to completed dwelling. A developer identifies an opportunity (perhaps acquiring an older single-family property in an established neighborhood). They must amplify this opportunity across distinct zones of operation, each with its own logic and gatekeepers.

First, the zone of finance: the developer must translate their vision into forms legible to lenders—pro formas, comparable sales, risk assessments. Different lenders have different appetites for different projects; financing for a \$600,000 single-family home comes from different sources than financing for four \$350,000 townhouses on the same parcel. The developer must know which banks or private investors fund which typologies and at what rates. This knowledge is tacit, earned through relationships and experience.

Second, the zone of regulation: the developer must navigate permitting processes, building codes, site plan approvals. Even after blanket rezoning, each project requires specific permits reflecting its particular details. The city's planning department may have removed one barrier (zoning restrictions) but didn't eliminate the need for detailed engagement with regulatory requirements. A developer experienced in single-family permitting doesn't automatically know the paths for townhouse approvals—different departments, different standards, different expertise required.

Third, the zone of labor: the developer must coordinate contractors, trades, and suppliers. Single-family construction operates through established networks—a developer knows which framing contractor they call, which electrician, which supplier delivers materials when needed. Transitioning to row housing or multiplexes means finding contractors with different capabilities, managing different construction sequences, coordinating more complex utility installations. Some contractors are unionized with formal procedures; others are informal operators paid in cash working on verbal agreements. The network is heterogeneous and relationship-dependent.

Each zone operates independently. A bank doesn't care whether the city permits the project—that's the developer's problem. The city doesn't care whether the developer secured financing—that's their responsibility. Contractors don't care about either—they want clear plans and reliable payment. The developer's task is ampliation: taking a concept from one zone (the initial idea) and successfully instantiating it across all necessary zones simultaneously. A viable project is one that exists across all these domains at once—financially viable AND regulatorily approved AND constructible with available labor.

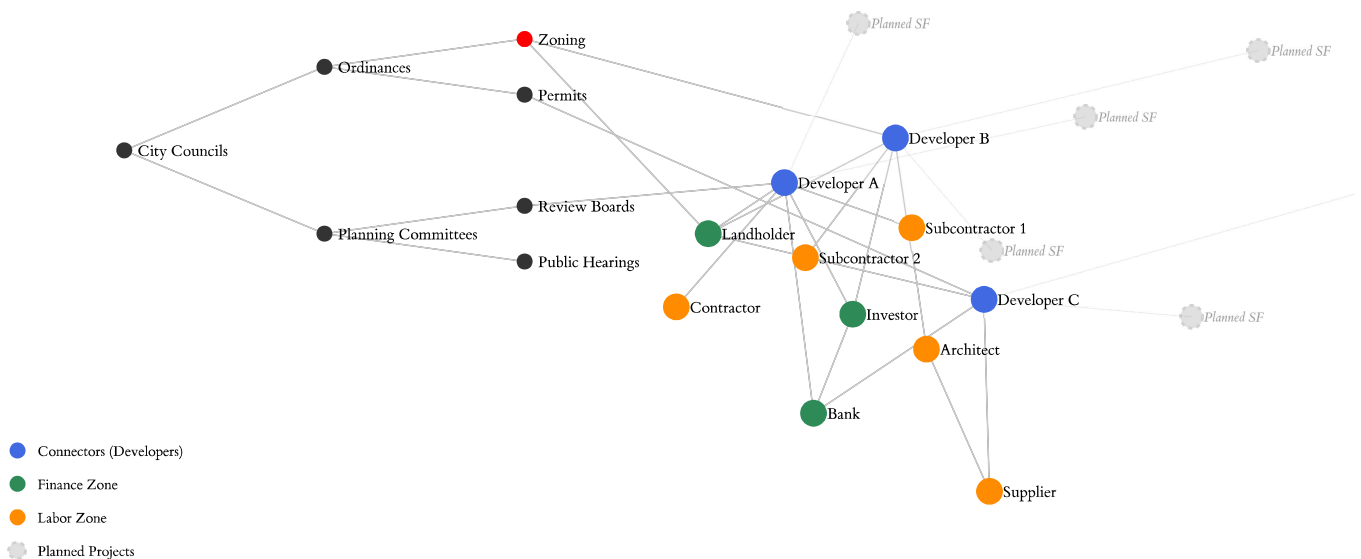
This is contingentist logic. The set of viable projects isn't fixed—it depends on which connections can be forged across zones at any given moment. When Calgary's bylaw passed, thousands of planned single-family projects became non-denoting terms. The phrase "the Smith family's planned bungalow on lot 4275" suddenly referred to nothing buildable under new rules. Plans that were progressing through financing became invalid. Permits in process became obsolete. Contractor agreements became void. These weren't adjustments to existing plans—they were ontological eliminations. Projects ceased to exist.

The recovery period we observe in the data—the lag causing the paradox—represents the slow work of building new connections. Developers must learn which lenders finance townhouses. Lenders must develop new risk models for denser typologies. Planners must establish new permit pathways. Contractors must acquire new capabilities or enter the market from other cities. This isn't adaptation of existing infrastructure—it's reconstruction of the network itself, connection by connection, relationship by relationship. The work is rhizomatic: distributed, bottom-up, without central coordination. It proceeds at the pace relationships form and knowledge diffuses, not at the pace bylaws change.

The Collision: When Arborescent Commands Meet Contingent Realities

Arborescent Logic: City Councils

Rhizomatic Logic: Construction Network



We can model this collision formally using concepts from modal logic. Think of the housing construction system as a hypersequent—a structure tracking truth across multiple parallel contexts simultaneously. A successful housing project must satisfy conditions across several domains: [financially viable in the capital

market | regulatorily approved in the city's system | constructible with available labor | marketable to eventual buyers]. Each zone has its own constraints and logics. A project succeeds only when all zones align.

The city council's bylaw changed one zone: regulatory requirements. But it didn't and couldn't change the others directly. Financial institutions still had loan products designed for single-family construction, not townhouse projects. Contractors still had crews trained for single-family techniques. Buyers still had expectations shaped by existing housing stock. The bylaw created immediate misalignment—projects that were viable in the regulatory zone became non-viable in financial and labor zones because the infrastructure connecting these zones hadn't adjusted.

This explains the statistical paradox. Construction didn't decline because rezoning was bad policy or because developers opposed density. Construction declined because the network infrastructure enabling construction was optimized for the old regulatory environment and became temporarily incompatible with the new one. The gap between actual construction (2,124 units/month) and trend projection (2,531 units/month) measures the destruction of network capacity faster than new capacity could form.

The compositional shift—single-family up 2.3%, multi-family up 15.9%—shows the network beginning to rebuild around new parameters. Some developers successfully forged new connections, securing financing for denser projects and finding capable contractors. But the substitution was insufficient because building rhizomatic networks takes time. You cannot command relationships into existence. You cannot decree that lenders will fund townhouses or that contractors will acquire new skills. These capabilities emerge through distributed experimentation—some developers succeed, others learn from them, knowledge diffuses gradually through the network. This is ampliation at network scale, and it operates on organic rather than bureaucratic timelines.

Implications for Policy Design: Working with Rhizomatic Structures

The lesson isn't that blanket rezoning was wrong. Removing artificial restrictions on density remains sound policy—Calgary's zoning code was genuinely restrictive, preventing construction of housing types many people want and need. The lesson is that hierarchical interventions into rhizomatic systems produce transition costs proportional to the intervention's scope and speed. Calgary changed every residential parcel simultaneously and immediately. The corresponding disruption was massive.

An alternative approach would recognize the rhizomatic nature of the construction market and design interventions to work with rather than against that structure. Instead of citywide instant implementation, phase the rezoning over 36 months across different neighborhoods. Early neighborhoods serve as laboratories—developers experiment with new typologies, lenders develop financing products, contractors build capabilities, and knowledge diffuses through the network. Later neighborhoods benefit from established infrastructure. The total timeline is longer, but construction momentum never breaks.

Or maintain blanket rezoning but provide transition support acknowledging destroyed network capacity. Establish a city loan program funding the delta between old and new typologies—bridging finance for developers converting single-family projects to townhouses. Create fast-track permitting for first-movers building denser housing, reducing regulatory uncertainty. Fund training programs for contractors learning new construction techniques. These aren't subsidies in the traditional sense—they're infrastructure investments recognizing that the city's regulatory change imposed real costs on market participants who must now rebuild their operational networks.

The deeper point is epistemological. Policymakers operating from arborescent positions—elected councils with legitimate authority to legislate—must resist the temptation to assume the systems they govern share their

hierarchical logic. Most complex social systems are rhizomatic: markets, ecosystems, cultural practices, informal institutions. These systems don't respond to commands the way machines respond to inputs. They respond through distributed adaptation as individual nodes forge new connections in response to changed environments. This adaptation takes time, follows bottom-up rather than top-down paths, and succeeds through experimentation and knowledge diffusion rather than through compliance with decrees.

If we want large-scale transitions—moving from postwar car-dependent sprawl to dense walkable urbanism, shifting from fossil fuel economies to renewable energy systems, transforming extractive industries to regenerative practices—we cannot simply legislate the end state and expect systems to adapt instantly. We must design interventions that acknowledge the rhizomatic structure of the systems we're trying to change and work with their internal logics. Provide time for networks to reconfigure. Support early adopters who bear transition costs. Create conditions enabling distributed experimentation. Allow knowledge to diffuse through existing channels rather than trying to impose it from above.

The Calgary rezoning paradox is a case study in what happens when we ignore this principle. A well-intentioned policy, correctly identifying an artificial constraint on housing construction, produced temporary reduction in construction because it treated a rhizomatic system as if it were a machine to be reprogrammed. The construction market will eventually adapt—new connections will form, new expertise will develop, new infrastructure will emerge. But the transition cost, measured in hundreds of foregone housing units during a housing crisis, represents the price of governing rhizomes with arborescent logic. We can do better by designing policies that respect the ontologies of the systems they aim to change.